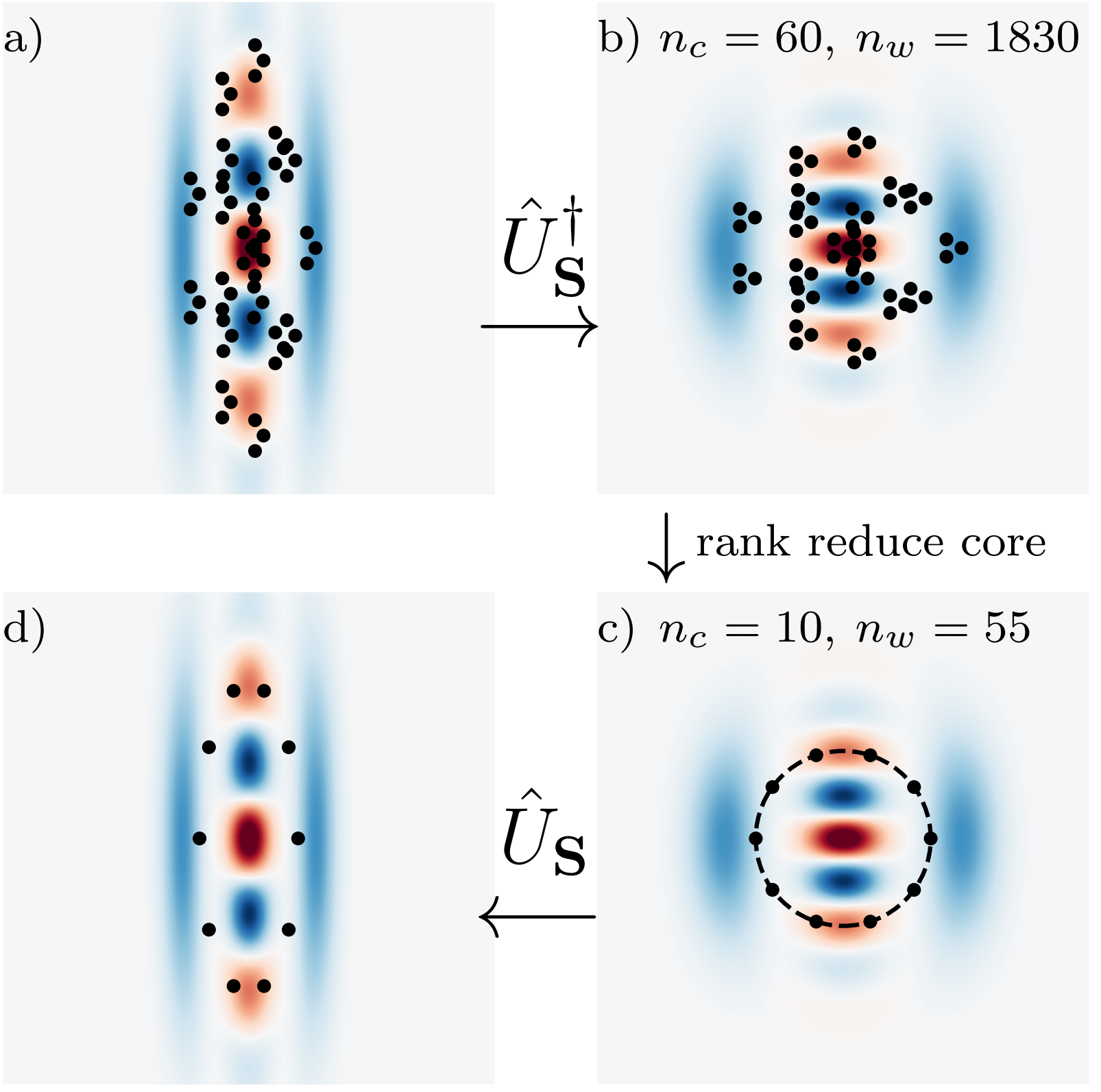

a)
b) $n_c = 60,\ n_w = 1830$
$\hat{U}_{\mathbf{S}}^{\dagger}$
rank reduce core
c) $n_c = 10,\ n_w = 55$
$\hat{U}_{\mathbf{S}}$
d)